

The Modern Software Developer

CS146S
Stanford University, Fall 2026
Mihail Eric

State of the World: 2026

AI • ANTHROPIC

Top engineers at Anthropic, OpenAI say AI now writes 100% of their code—with big implications for the future of software development jobs

By Beatrice Nolan
Tech Reporter

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January 29, 2026, 2:47 PM ET



BUSINESS INSIDER

EXCLUSIVE

Meta is pushing employees to use AI, and this doc shows how much

BUSINESS INSIDER

CULTURE

This '90s stoner trend is now the latest Gen Z obsession: the hacky sack



What That Means For You

- Broader industry-wide AI adoption means companies **need** you
 - Becoming “AI-native” is now the new goal
- But...the floor is higher
- Software developers are more productive than they have ever been in history and so expectations have grown
- Focus on holistic systems-level thinking, architectures, and abstractions
 - When combined with AI-first development practices you will be irreplaceable

The Modern Software Developer

This class is about principles not tools

10 weeks in 1 slide

The Takeaway

- Human-agent engineering
 - Focus on the skills that are not yet replaced by AI systems
 - Business and product understanding
 - Become the tech architect – develop good taste for software systems
 - Communication is crucial
- LLMs are only as good as you are
 - Good context leads to good code
 - If you can't understand your codebase, neither will an LLM
 - Prevent AI slop
- Experiment aggressively

Course Logistics

- Instructor (Hi!)

- Stanford undergrad/grad
- Head of AI at Monaco, an SF-based startup in the sales space
- AI Advisor for dozens of companies from seed-stage startups to public enterprises
- Built first LLMs at Amazon Alexa
- Founded and sold an ML education startup
- Founded a YC-backed AI coding company



- 2 awesome CAs

- Isaac Kan
- Vijay Daita



Course Logistics

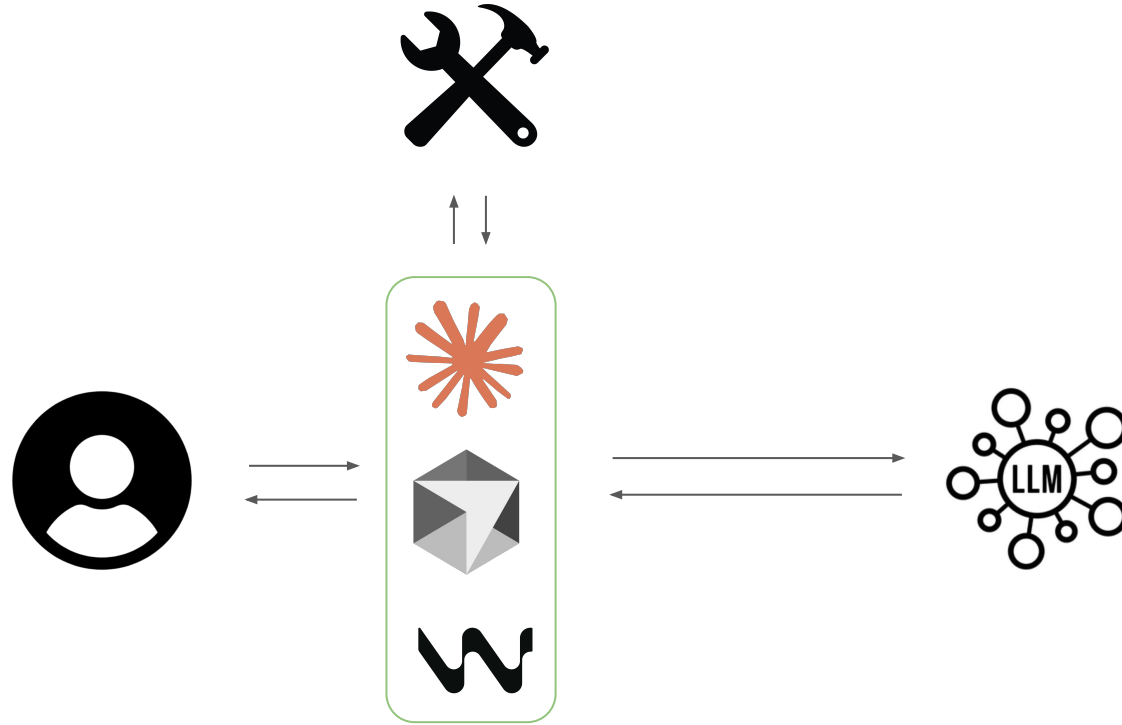
- <https://themodernsoftware.dev>
- Lectures
 - Tue/Thur 5:30-6:20 pm
- Deliverables
 - 6 assignments (1x/week) focusing on lecture material practice
 - <https://github.com/mihail911/modern-software-dev-assignments>
 - 3-4 contributions to OSS repositories
 - 1 final open-ended project in which you will exercise AI coding principles we cover
- Grading
 - 50/15/30/5 breakdown for project/assignments/OSS/participation

Course Logistics

- OSS Partners
 - 15 of the top AI-native open-source projects including Vercel, Warp, Marimo, Milvus, Pi, CrewAI, Browserbase, HeyGen, CopilotKit, Semgrep, OpenHands, Cmux, Arize, Sloth, Anyscale
- Guest Lectures
 - Founders and engineering execs leading top AI developer teams today (creators of Claude Code, Devin, Replit, Factory, and more)
 - Billions of dollars raised, tens of billions in valuation
 - Don't miss these talks!

How to Build Claude Code in 200 Lines of Code

It's that simple



Steps

- Read in terminal and keep appending to conversation
- Tell LLM what tools are available
 - It asks for tool use at appropriate time
 - You execute tool offline and return response
 - Tools
 - “Read_file”
 - “List_dir”
 - “Edit_file”

Let's build a coding agent from scratch!



Questions?